

THALES



payShield 9000

payShield 9000 and AES
Application Note

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>> References

Some of the documents referred to below are confidential and have limited availability.

No.	Document	Publisher
1	1270A546 payShield 9000 Host Command Reference Manual. (<i>This document has restricted availability.</i>)	Thales
2	1270A593 payShield 9000 General Information Manual. (<i>This document has restricted availability.</i>)	Thales
3	1270A544 payShield 9000 Console reference Manual. (<i>This document has restricted availability.</i>)	Thales
4	1270A596 Local HSM Manager's User Guide. (<i>This document has restricted availability.</i>)	Thales
5	PWPRO530 payShield 9000 Application Note - Migrating between LMKs	Thales

>> Introduction

The algorithm originally used to encrypt card payment data was DES, subsequently replaced by TDES. Virtually all payments data encryption is currently based on TDES (with DES still being used in some areas).

AES is (like DES and TDES) a symmetric algorithm used to encrypt data, but provides significantly higher encryption strength than DES/TDES. AES keys have lengths of 128, 192, or 256 bits. The comparative strengths of the algorithms are:

Algorithm	Key Length	Bits of Security
DES	56-bit	56
TDES	Double-length (112-bit)	80
TDES	Triple-length (168-bit)	112
AES	128-bit	128
AES	192-bit	192
AES	256-bit	256

AES is currently not widely used in the payments industry, but it is beginning to be used, and its use is expected to become widespread. EMVCo, who define specifications for payment smartcards, are planning to introduce AES into their testing environment as an alternative to TDES, and various standards bodies are working on an agreed specification for AES PIN Blocks. Various solutions for Point-to-Point Encryption (P2PE) and Format Preserving Encryption (FPE) make use of AES.

Support for AES in the payShield 9000 allows users to prepare for the future of payments processing and card issuing as well as participating in new applications such as P2PE and FPE.

This document describes AES support in payShield 9000 v2.0 and v2.1.

>> The Thales payShield 9000

The payShield 9000 is the latest model of payment HSMs (Hardware Security Modules) from Thales. (It is backwards compatible with its predecessor, the HSM 8000, but the HSM 8000 provides no support for AES and therefore this Application Note does not apply to that model.)

The payShield 9000 is an HSM that is designed specifically for use in processing card payments and in associated activities such as card issuance. Its functions are geared up to payment-related activities (such as PIN Block processing, or generating and verifying card verification values).

The HSM is essentially a peripheral to a host computer system which is running a payment-related application, and provides cryptographic functions in a protected, secure environment. It meets the appropriate standards and has the required certifications. Use of an HSM is advised as best practise for security, and is mandated for certain types of application.

payShield 9000 Host Commands

The way that the host system and HSM exchange data is by exchanging messages: the host sends a command to the HSM, and the HSM sends a response. The payShield 9000 does not require any software to be installed on the host, and can be used by any host which supports one of the following communication methods:

- Ethernet - TCP/IP
- Ethernet - UDP
- FICON (IBM proprietary fiber-optic networking)
- Asynchronous serial

The commands sent by the host are identified by a 2-character identifier (e.g. "A2", "GK"), and the content of each command (and its response) is different to reflect the different parameters and output fields required. In this document, the commands are referred to by their 2-character identifier and information is provided about how these commands have been modified to support AES: the detailed command structure is provided in [*Reference 1*](#).

Examples of host command structures, for both Variant and Keyblock LMKs (see [*below*](#)) are given in [*Appendix B*](#).

The payShield 9000 Console

The traditional method of managing Thales payment HSMs was by the use of a console - an 80 line x 24 character text display running over a serial asynchronous communication link, or "dumb terminal". This type of terminal is now seen as old-fashioned, but is nevertheless one of the methods supported on the payShield 9000 to provide operational compatibility with older HSMs being used in mature implementations.

The console is used by entering command codes consisting of one or more alphanumeric characters to initiate a command, and then entering parameters specific to the desired command in response to prompts.

This document identifies how the commands have been modified to support AES: the detailed operation of the commands can be found in [Reference 3](#).

Local HSM Manager

Local HSM Manager is a more modern method of managing the payShield 9000. It consists of a Linux application loaded onto a standard PC hardware platform connected to the payShield 9000 using a point-to-point Ethernet connection.

It is designed as a direct replacement for the console (and must be located next to the payShield 9000), offering a more modern graphical user interface than the console.

Local HSM Manager is used by selecting options from menus and then entering data into dialog boxes. This document explains how these options have been modified for AES. Detailed operation of Local HSM Manager is described in [Reference 4](#).

Remote HSM Manager

Remote HSM Manager offers a similar user interface to that provided by Local HSM Manager, but allows the payShield 9000 to be managed remotely over a TCP/IP link from anywhere in the world.

Current versions of Remote HSM Manager do not provide support for AES, and so this document does not discuss its use.

Local Master Keys (LMKs)

LMKs are the critical secrets on the payShield 9000, and are automatically deleted if an attempt is made to attack the HSM. The LMKs are used to encrypt all operational keys (including other master keys).

The payShield 9000 supports 2 types of LMK:

- **Variant LMK:** this is the traditional type of LMK and still in most widespread use. Key separation is provided by creating a variant of the LMK to encrypt different types of key or data.
- **Keyblock LMK:** this is a more modern and more secure type of LMK. Keys encrypted using the LMK are incorporated into a keyblock which includes metadata defining aspects such as exportability of the key and what it can be used for. The keyblock structure used for this is a proprietary Thales design, based on the TR-31 keyblock described below.

Variant and Keyblock LMKs are described in [Reference 2](#). Examples of host commands using each type of LMK are given in [Appendix B](#).

A single payShield 9000 can have multiple LMKs installed, with each LMK managed by a different security team. This allows multiple applications or clients to be securely segregated on the same hardware unit. The LMKs installed on a payShield 9000 can be a mix of variant and keyblock types, and DES/TDES and AES algorithms.

AES Keyblock LMKs must be used to protect AES keys. Such AES keyblock LMKs can also be used to protect any other keys or data, such as double- or triple-length TDES or RSA keys. AES Variant LMKs are not supported.

Use of Keyblocks for exporting keys

An important capability of the payShield 9000 is its support for TR-31 keyblocks when exporting keys.

Traditionally, key management specified in ANSI X9.17 has been used to exchange keys in a multi-vendor environment, where keys are exchanged between products sourced from different manufacturers such that proprietary exchange mechanisms cannot be used.

However, X9.17 key exchange has a number of well-recognised security deficiencies, such as being unable to prevent attacks based on key masquerading and not providing mechanisms for detecting modifications made to the transmitted key.

The deficiencies in X9.17 are addressed by the use of TR-31 keyblocks for key exchange. The TR-31 specification (produced by the ANSI X9 committee) requires the key being exchanged to be formatted within a block which includes restrictions on how that key can be used and managed, and uses a MAC to prevent modifications to the data from going undetected.

The payShield 9000 host commands referenced in this document support the use of TR-31 key blocks as well as the old X9.17 method. Use of X9.17 is important for support of legacy products, but developers of new card payment hardware products should consider the use of TR-31 keyblocks for receiving keys as an opportunity to enhance the security of their systems.

>> Summary of AES support in the payShield 9000

Support for the AES algorithm in payShield 9000 v2.0 and v2.1 includes:

- 256-bit AES Keyblock LMKs to protect AES working keys plus other keys (except single DES) and data protected in the HSM by LMKs.
- Use of AES for ZMKs and TMKs.
- Use of AES working keys for data encryption and MACing.
- Support for 128-, 192-, and 256-bit AES keys.
- Export/import of AES working keys using Thales Keyblocks. (Note: the TR-31 standard does not currently support the AES algorithm.)
- Use of a Console or Local HSM Manager to manage AES keys and AES Keyblock LMKs.

The next chapter gives information on implementing AES on existing payShield 9000s.

Later chapters indicate how host commands, console commands, and Local HSM Manager menu options have been modified to support AES.

The following table shows the keys used in the payShield 9000, indicating which keys can use AES.

Key Description	AES*	Keyblock LMK Key Usage	Variant LMK Key Type ^a	Key Type Code
Base Derivation Key (BDK)		B0	009	BDK
Base Derivation Key (BDK) Type 2		41	609	BDK2
Base Derivation Key (BDK) Type 3		42	809	BDK3
Card Verification Key		C0	402	CVK
Card Verification Key (American Express CSC)		11	402	CSCK
Card Verification Key (Mastercard CVC)		12	402	CVK
Card Verification Key (Visa CVV)		13	402	CVK
CBC MAC (AES)	Yes	M5	n/a	n/a
CMAC (AES)	Yes	M6	n/a	n/a
Data Encryption Key	Yes	21	00B	DEK
Data Encryption Key (Generic)	Yes	D0	00B, 00A, 30B	DEK, ZEK, TEK

Key Description	AES *	Keyblock LMK Key Usage	Variant LMK Key Type ^a	Key Type Code
Dynamic CW Master Key		32	709	MK-CVC3
EMV/Chip card Master Key: Application Cryptogram		E0	109	MK-AC
EMV/Chip card Master Key: Data Authentication Code		E3	409	MK-DAC
EMV/Chip card Master Key: Dynamic Numbers		E4	509	MK-DN
EMV/Chip card Master Key: Secure Messaging for Confidentiality		E1	309	MK-SMC
EMV/Chip card Master Key: Secure Messaging for Integrity		E2	209	MK-SMI
HMAC key (using SHA-1)		61	10C	HMAC
HMAC key (using SHA-224)		62	10C	HMAC
HMAC key (using SHA-256)		63	10C	HMAC
HMAC key (using SHA-384)		64	10C	HMAC
HMAC key (using SHA-512)		65	10C	HMAC
IPEK (DUKPT)		B1	302	IPEK
ISO 16609 MAC algorithm 1 (using 3-DES)		M0	003, 008	TAK, ZAK
ISO 9797-1 MAC algorithm 1		M1	003, 008	TAK, ZAK
ISO 9797-1 MAC algorithm 3		M3	003, 008	TAK, ZAK
Key Encryption / Wrapping Key (Generic)	Yes	K0	002 / 80D, 000	TMK, ZMK
PIN Encryption Key (Generic)		P0	002 / 70D	PEK
PIN Verification Key (Generic)		V0	002	PVK
PIN Verification Key (IBM 3624 algorithm)		V1	002	PVK
PIN Verification Key (Visa PVV algorithm)		V2	002	PVK

Key Description	AES *	Keyblock LMK Key Usage	Variant LMK Key Type ^a	Key Type Code
RSA Private Key		03	00C	RSA-SK
RSA Public Key		02	00D	RSA-PK
Terminal Key Encryption Key	Yes	51	002 / 80D	TMK
Terminal PIN Encryption Key		71	002 / 70D	TPK
Terminal/Data Encryption Key (TEK)	Yes	23	30B	TEK
Transaction Key Scheme Terminal Key Register		73	002 / 90D	TKR
Visa Cash Master Load Key (KML)			200	KML
WatchWord Key		01	006	WWK
Zone Key Encryption Key	Yes	52	000	ZMK
Zone PIN Encryption Key		72	001	ZPK
Zone/Data Encryption Key	Yes	22	00A	ZEK

* Only available when using an AES Keyblock LMK

^a Where 2 key types are given (e.g. 002/70D), the key type depends on the security setting “*Enforce key type 002 separation for PCI HSM compliance*” - see Chapter 10 in [Reference 2](#).

This table does not include keys used for AS2805 (optional license LIC003), OBKM (optional license LIC004), and card issuing (optional licenses O11, O16, O18, O23). The commands used in these licenses do not support keyblock LMKs and therefore cannot use AES keys.

As and when updates to the standards published by bodies such as ANSI X9, ISO, and EMVCo become available, the payShield 9000 software will be extended to support additional AES key usages.

Support of AES requires optional license HSM9-LIC007 to be installed on the payShield 9000.

>> Implementing AES on an existing payShield 9000

Re-encrypting existing keys under an AES Keyblock LMK

An AES Keyblock LMK must be installed if it is required to use AES working keys: these keys cannot be protected using a Variant or Keyblock TDES LMK.

Even if there is no requirement to use AES working keys, users may wish to use AES Keyblock LMKs to protect TDES and other working keys because of the greater security strength that the 256-bit AES Keyblock LMK provides compared to a TDES LMK.

Where a payShield 9000 is equipped with an appropriate number of LMKs, it may not be necessary to re-encrypt existing working keys under a new AES Keyblock LMK: the existing working keys could be left encrypted under the existing LMKs, with only new working keys (including AES keys) being encrypted under a new AES Keyblock LMK.

The multiple LMK approach will not be satisfactory - for example if:

- the payShield 9000 has an insufficient number of LMKs available,
- only one LMK can be used because the host applications have not been adapted to make use of multiple LMKs,
- the AES Keyblock LMK is being used to provide enhanced security for existing working keys and data.

In this case, it will be necessary to re-encrypt existing working keys and data which are still required from the old LMK that they are currently encrypted under to the new AES LMK.

The process for such an LMK migration is described in [*Reference 5*](#). This process is based on:

- multiple uses of the *BW* host command, which re-encrypts a working key from an old LMK to a new LMK,
- multiple uses of the *BG* host command to re-encrypt PINs,
- multiple use of the *LO* host command to re-encrypt decimalization tables.

[*Reference 5*](#) also outlines how to install the AES Keyblock LMK.

Note that:

- Any type of working key (including key encryption keys, but excluding single-DES keys) and any data that can be encrypted under a payShield 9000 LMK can be encrypted using an AES Keyblock LMK. The use of AES Keyblock LMKs is not limited to AES keys.
- AES keys cannot be encrypted under a TDES LMK (whether Variant or Keyblock type).
- Keys can be re-encrypted from any type of LMK to an AES Keyblock LMK.
- Keys encrypted under an AES Keyblock LMK can only be re-encrypted to another AES Keyblock LMK - they cannot be migrated to a TDES LMK of any type.

Creating AES working keys

Once an AES Keyblock LMK has been installed, it is possible to create (i.e. generate or import) AES working keys, encrypted under the AES Keyblock LMK, for use with the payShield 9000. It may also be necessary to export AES working keys.

This can be achieved with the key management facilities available using:

- Console commands - see the section on *AES Key Management* in the chapter on changes made to console command.
- Host commands - see the section on *AES Key Management* in the chapter on changes made to host commands.
- Local HSM Manager - see the section on *AES Key Management* in the chapter on changes made to Local HSM Manager menu options.

>> Changes to host commands to support AES

Processing data using AES keys

The commands in this table allow AES working keys to be used to process data. These commands also support the use of AES keyblock LMKs.

Host Command	Summary of Changes Made
M0 Encrypt Data Block	Support for AES encryption keys.
M2 Decrypt Data Block	Support for AES encryption keys.
M4 Translate Data Block	Support for Source/Destination AES encryption keys.
M6 Generate MAC	Choice of MAC Algorithm extended to include: '5' : CBC-MAC (AES only) '6' : CMAC (AES only) Choice of Padding Method extended to include: '4' : AES CMAC Padding. Key can be a CMAC or CBC-MAC key.
M8 Verify MAC	Choice of MAC Algorithm extended to include: '5' : CBC-MAC (AES only) '6' : CMAC (AES only) Choice of Padding Method extended to include: '4' : AES CMAC Padding. Key can be a CMAC or CBC-MAC key.
MY Verify and Translate MAC	Choice of Source/Destination MAC Algorithm extended to include: '5' : CBC-MAC (AES only) '6' : CMAC (AES only) Choice of Source/Destination Padding Method extended to include: '4' : AES CMAC Padding. Source/Destination Key can be a CMAC or CBC-MAC key.

AES Key Management

The following key management host commands can be used with AES working keys. These commands also support AES keyblock LMKs.

Host Command	Summary of Changes Made
A0 Generate a key	Choice of Algorithm when using AES keyblock LMKs extended to include: 'A1' – 128-bit AES key 'A2' – 192-bit AES key 'A3' – 256-bit AES key Generated key can be encrypted under an AES ZMK or TMK. AES keys can also be encrypted using a TDES ZMK/TMK.
A2 Generate and Print a Component	Choice of Algorithm when using AES keyblock LMKs extended to include: 'A1' – 128-bit AES key 'A2' – 192-bit AES key 'A3' – 256-bit AES key
A4 Form a Key from Encrypted Components	Allows an AES key to be formed from components and encrypted under an AES Keyblock LMK.
A6 Import a Key	Allows an AES key to be imported. Support is provided for AES ZMKs, but the AES key can also be encrypted using a TDES ZMK.
A8 Export a Key	An AES key can be encrypted under a ZMK/TMK. Support is provided for AES ZMK/TMKs, but the AES key can also be encrypted using a TDES ZMK/TMK.
BY Translate ZMK from ZMK to LMK encryption	The ZMK being translated can be an AES ZMK. And the encrypting ZMK can be an AES key. The LMK that the ZMK is to be translated to must, of course, be an AES Keyblock LMK.
BW Translate Keys from Old LMK to New LMK and Migrate to New Key Type	Supports AES keys & LMKs
NE Generate and Print a Key as Split Components	Choice of Algorithm when using AES keyblock LMKs extended to include: 'A1' – 128-bit AES key 'A2' – 192-bit AES key 'A3' – 256-bit AES key

AES Keyblock LMK support

The following host commands are able to accept keys and other data protected by AES keyblock LMKs.

Host Command		Summary of Changes Made
BA	Encrypt a Clear PIN	The output PIN can be encrypted using an AES Keyblock LMK.
BC	Verify a Terminal PIN Using the Comparison Method	The input PIN can be encrypted using an AES Keyblock LMK.
BE	Verify an Interchange PIN Using the Comparison Method	The input PIN can be encrypted using an AES Keyblock LMK.
BG	Translate a PIN and PIN Length	The input PIN can be encrypted under an AES Keyblock LMK.
BK	Generate an IBM PIN Offset (of a customer selected PIN)	Support is provided for decimalization tables encrypted using AES Keyblock LMKs. (NOTE: AES PIN Block formats are not defined, so the input PIN Block must be DES-encrypted.)
BQ	Translate PIN Algorithm	PIN can be encrypted using an AES Keyblock LMK. ??? confirm - PINs are encrypted using the Visa & Racal algorithms ???
BS	Erase the Key Change Storage	Supports AES keys
BU	Generate a Key Check Value	Supports AES keys
CA	Translate a PIN from TPK to ZPK Encryption	Supports TDES TPK/ZPK encrypted under an AES Keyblock LMK
CC	Translate a PIN from One ZPK to Another	Supports TDES ZPK encrypted under an AES Keyblock LMK
CE	Generate a Diebold PIN Offset	The input PIN can be encrypted using an AES Keyblock LMK.
CG	Verify a Terminal PIN Using the Diebold Method	Supports TDES TPK encrypted under an AES Keyblock LMK
CI	Translate a PIN from BDK to ZPK Encryption (DUKPT)	Supports TDES BDK/ZPK encrypted under an AES Keyblock LMK
CK	Verify a PIN Using the IBM Method (DUKPT)	Support is provided for decimalization tables encrypted using AES Keyblock LMKs.
CM	Verify a PIN Using the VISA PVV Method (DUKPT)	Supports TDES BDK/PVK encrypted under an AES Keyblock LMK

Host Command		Summary of Changes Made
CO	Verify a PIN Using the Diebold Method (DUKPT)	Supports TDES BDK encrypted under an AES Keyblock LMK
CQ	Verify a PIN Using the Encrypted PIN Method (DUKPT)	The encrypted data base PIN can be encrypted using AES Keyblock LMKs.
CS	Modify Keyblock Header	Supports AES Keyblock LMKs
CU	Verify & Generate a VISA PVV (of a customer selected PIN)	Supports TDES TPK/ZPK/PVK encrypted under an AES Keyblock LMK
CW	Generate a Card Verification Code/Value	Supports TDES CVK encrypted under an AES Keyblock LMK
CY	Verify a Card Verification Code/Value	Supports TDES CVK encrypted under an AES Keyblock LMK
DA	Verify a Terminal PIN Using the IBM Method	Support is provided for decimalization tables encrypted using AES Keyblock LMKs.
DC	Verify a Terminal PIN Using the VISA Method	Supports TDES TPK/PVK encrypted under an AES Keyblock LMK
DE	Generate an IBM PIN Offset (of an LMK encrypted PIN)	Support is provided for decimalization tables encrypted using AES Keyblock LMKs. The input PIN can be encrypted using an AES Keyblock LMK.
DG	Generate a VISA PIN Verification Value (of an LMK encrypted PIN)	The input PIN can be encrypted using an AES Keyblock LMK.
DU	Verify & Generate an IBM PIN Offset (of customer selected new PIN)	Support is provided for decimalization tables encrypted using AES Keyblock LMKs.
EA	Verify an Interchange PIN Using the IBM Method	Support is provided for decimalization tables encrypted using AES Keyblock LMKs.
EC	Verify an Interchange PIN Using the VISA Method	Supports TDES ZPK/PVK encrypted under an AES Keyblock LMK
EE	Derive a PIN Using the IBM Method	Support is provided for decimalization tables encrypted using AES Keyblock LMKs. The generated PIN can be encrypted using an AES Keyblock LMK.
EG	Verify an Interchange PIN Using the Diebold Method	Supports TDES ZPK encrypted under an AES Keyblock LMK
EI	Generate a Public/Private Key Pair	Output private key can encrypted under an AES Keyblock LMK.

Host Command		Summary of Changes Made
EK	Load a Private Key	Input private key can encrypted under an AES Keyblock LMK.
EM	Translate a Private Key	Input & Output private key can encrypted under an AES Keyblock LMK.
EO	Import a Public Key	MAC using AES Keyblock LMK
EQ	Validate a Public Key	MAC using AES Keyblock LMK
ES	Validate a Certificate and Import the Public Key	??? Is the input & output public key encrypted - so is the LMK type relevant ??? The input private key can be encrypted using an AES Keyblock LMK.
EU	Translate a Public Key	MAC using AES Keyblock LMK
EW	Generate an RSA Signature	Input private key can encrypted under an AES Keyblock LMK.
EY	Validate an RSA Signature	MAC using AES Keyblock LMK
FU	Verify a Watchword Response	Supports TDES WWK encrypted under an AES Keyblock LMK
FW	Generate a VISA PIN Verification Value (of a customer selected PIN)	Supports TDES TPK/ZPK/PVK encrypted under an AES Keyblock LMK
G0	Translate a PIN from BDK to ZPK Encryption (3 DES DUKPT)	The input BDK and ZPK can be encrypted using an AES Keyblock LMK.
GA	Derive a PIN using the Diebold method	The generated PIN can be encrypted using an AES Keyblock LMK.
GI	Import Key under an RSA Public Key	??? Is the input public key encrypted - so is the LMK type relevant ??? The imported private key can be encrypted under an AES Keyblock LMK. The output TDES key can be encrypted using an AES keyblock LMK.
GK	Export Key under an RSA Public Key	??? Is the input public key encrypted - so is the LMK type relevant ??? The input private, TDES, and HMAC keys can be encrypted using an AES Keyblock LMK.
GO	Verify a PIN Using the IBM Method (3 DES DUKPT)	Support is provided for decimalization tables encrypted using AES Keyblock LMKs.

Host Command		Summary of Changes Made
GQ	Verify a PIN Using the VISA PVV Method (3 DES DUKPT)	Supports TDES BDK/PVK encrypted under an AES Keyblock LMK
GS	Verify a PIN Using the Diebold Method (3 DES DUKPT)	Supports TDES BDK encrypted under an AES Keyblock LMK
GU	Verify a PIN Using the Encrypted PIN Method (3 DES DUKPT)	The encrypted data base PIN can be encrypted using AES Keyblock LMKs.
GW	Generate/Verify a MAC (3 DES DUKPT)	Supports TDES BDK encrypted under an AES Keyblock LMK
JA	Generate a Random PIN	The generated PIN can be encrypted using an AES Keyblock LMK.
JC	Translate a PIN from TPK to LMK Encryption	The output PIN can be encrypted using an AES Keyblock LMK. (NOTE: AES PIN Block formats are not defined, so the input PIN Block must be DES-encrypted.)
JE	Translate a PIN from ZPK to LMK Encryption	The output PIN can be encrypted using an AES Keyblock LMK.
JG	Translate a PIN from LMK to ZPK Encryption	The input PIN can be encrypted using an AES Keyblock LMK. (NOTE: AES PIN Block formats are not defined, so the input PIN Block must be DES-encrypted.)
K0	Decrypt Encrypted Counters (EMV 4.x)	Input TDES MK-AC key can be encrypted under AES Keyblock LMK
K2	Verify Truncated Application Cryptogram (MasterCard CAP)	Input TDES MK-AC key can be encrypted under AES Keyblock LMK
KQ	ARQC Verification and/or ARPC Generation (EMV 3.1.1)	Input TDES MK-AC key can be encrypted under AES Keyblock LMK
KS	Data Authentication Code and Dynamic Number Verification (EMV 3.1.1)	Input TDES MK-DAC and MK-DN keys can be encrypted under AES Keyblock LMK
KU	Generate Secure Message (EMV 3.1.1)	Input TDES MK-SMI, MK-SMC, TK, source PEK, and MK-AC keys can be encrypted under AES Keyblock LMK
KY	Generate Secure Message (EMV 4.x)	Input TDES MK-SMI, MK-SMC, TK, source PEK, and MK-AC keys can be encrypted under AES Keyblock LMK
KW	ARQC Verification and/or ARPC Generation (EMV 4.x)	Input TDES MK-AC key can be encrypted under AES Keyblock LMK

Host Command		Summary of Changes Made
L0	Generate an HMAC Secret Key	Output HMAC key can be encrypted under AES Keyblock LMK
LC	Verify the Diebold Table in User Storage	The Diebold table can be encrypted using an AES Keyblock LMK.
LK	Generate a Decimal MAC	Supports TDES TAK encrypted under an AES Keyblock LMK
LM	Verify a Decimal MAC	Supports TDES TAK encrypted under an AES Keyblock LMK
LO	Translate Decimalization Table from Old to New LMK	Support is provided for input/output decimalization tables encrypted using AES.
LQ	Generate an HMAC on a Block of Data	Input HMAC key can be encrypted under AES Keyblock LMK
LS	Verify an HMAC on a Block of Data	Input HMAC key can be encrypted under AES Keyblock LMK
LU	Import an HMAC key under a ZMK	Supports TDES ZMK encrypted under an AES Keyblock LMK. The output HMAC key can be encrypted under an AES Keyblock LMK.
LW	Export an HMAC key under a ZMK	Supports TDES ZMK encrypted under an AES Keyblock LMK. The input HMAC key can be encrypted under an AES Keyblock LMK.
LY	Translate a HMAC Key from Old LMK to New LMK	The old and new LMKs can be AES Keyblock LMKs. ??? Can you convert from AES to non-AES ???
NG	Decrypt an Encrypted PIN	The input PIN can be encrypted using an AES Keyblock LMK.
OA	Print a PIN Solicitation Mailer	Reference number check value can use an AES Keyblock LMK
PE	Print PIN/PIN and Solicitation Data	The input PIN can be encrypted using an AES Keyblock LMK.
PG	Verify PIN/PIN and Solicitation Mailer Cryptography	The input PIN can be encrypted using an AES Keyblock LMK.
PM	Verify a Dynamic CVV/CVC	Supports TDES MK-DCVV encrypted under an AES Keyblock LMK
Q0	Translate Audit Record MAC key	The old and new LMKs can be AES Keyblock LMKs. ??? Can you convert from AES to non-AES ???

Host Command		Summary of Changes Made
QA	Load Solicitation Data to User Storage	Reference number can be protected using an AES Keyblock LMK
QC	Final Load of Solicitation Data to User Storage	Reference number can be protected using an AES Keyblock LMK
RA	Cancel Authorized Activities	Authorized activities for an AES Keyblock LMK can be cancelled.
RC	Verify Solicitation Mailer Cryptography	Reference number check value can use an AES Keyblock LMK
RI	Transaction Request With a PIN (T/AQ Key)	The input Terminal Key Register can be encrypted using an AES Keyblock LMK. The output Response MAC Residue and TPK can be encrypted using an AES Keyblock LMK.
RK	Transaction Request Without a PIN	The input Terminal Key Register can be encrypted using an AES Keyblock LMK. The output Response MAC Residue can be encrypted using an AES Keyblock LMK.
RM	Administration Request Message	The input Terminal Key Register can be encrypted using an AES Keyblock LMK. The output Response MAC Residue can be encrypted using an AES Keyblock LMK.
RO	Transaction Response with Auth Para from Card Issuer	The input Terminal Key Register, ZPK, and Request MAC Residue can be encrypted using an AES Keyblock LMK. The output Response MAC Residue and Terminal Key Register can be encrypted using an AES Keyblock LMK.
RQ	Generate Auth Para and Transaction Response	The input Terminal Key Register and Request MAC Residue can be encrypted using an AES Keyblock LMK. The output Response MAC Residue and Terminal Key Register can be encrypted using an AES Keyblock LMK.
RS	Confirmation	The input Terminal Key Register and Request MAC Residue can be encrypted using an AES Keyblock LMK.
RU	Transaction Request With a PIN (T/CI Key)	Support for output KEYVAL encrypted under an AES Keyblock LMK.
RW	Translate KEYVAL	Support for input KEYVAL encrypted under an AES Keyblock LMK.
RY	(Mode 3) Calculate Card Security Codes	Support for input CSMK encrypted under an AES Keyblock LMK.



Host Command		Summary of Changes Made
RY	(Mode 4) Verify Card Security Codes	Support for input CSCK encrypted under an AES Keyblock LMK.

>> Changes to console commands to support AES

Managing the HSM

These console commands, used to manage the payShield 9000, have been modified to take account of the AES algorithm.

Console Command		Summary of Changes Made
DT	Diagnostic Test	Tests the AES algorithm
VR	View Software Revision Number	Now reports on availability of AES algorithm and relevant NIST certifications.

AES Key Management

The following key management console commands can be used with AES working keys. These commands also support AES keyblock LMKs.

Console Command		Summary of Changes Made
CK	Generate a Check Value	Can now generate a value for keys, including AES keys, encrypted under an AES keyblock LMK.
EC	Encrypt Clear Component	Can now encrypt an AES component. TDES components may be, and AES components must be, encrypted under an AES keyblock LMK.
FK	Form Key from Components	Can now form an AES key from components. TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK.
GC	Generate Key Component	Can now generate an AES key component. TDES components may be, and AES components must be, encrypted under an AES keyblock LMK.
GS	Generate Key and Write Components to Smartcard	Can now generate AES keys/components. TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK.
IK	Import Key	Can now import an AES key. The ZMK used to encrypt TDES or AES keys to be imported can be an AES key and/or can be encrypted under an AES keyblock LMK. Imported TDES keys may be, and imported AES keys must be, encrypted under an AES keyblock LMK.

Console Command	Summary of Changes Made
KE Export Key	Can now export an AES key. For the key to be exported, TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK. The ZMK used to encrypt the exported keys can be an AES key and/or can be encrypted under an AES keyblock LMK.
KG Generate Key	Can now generate an AES key. TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK.

AES Keyblock LMK support

The following console commands are able to accept keys and other data protected by AES keyblock LMKs.

Console Command	Summary of Changes Made
A Enter the Authorized State / Authorize Activity	Can now authorize for AES keyblock LMKs.
CO Create an Authorizing Officer Smartcard	Can now copy cards containing AES keyblock LMK components.
CV Generate a Card Verification Value	Can now use a CVK encrypted under an AES keyblock LMK.
DC Duplicate LMK Component Sets	Can now copy cards holding AES keyblock LMK components.
DM Delete LMK	Can now delete AES keyblock LMKs.
DO Delete 'Old' LMK from Key Change Storage	Can now delete AES keyblock LMKs.
ED Encrypt Decimalization Table	Decimalisation tables can now be encrypted under an AES keyblock LMK.
GK Generate LMK Component	Can now generate AES keyblock LMK components. A single use of the command generates all the required components.
LK Load LMK	Can now load AES keyblock LMKs.
LO Move 'Old' LMKs into Key Change Storage	Can now work with AES keyblock LMKs. AES Keyblock LMKs cannot be loaded as the old LMK where the new LMK is TDES (either variant or keyblock).

Console Command		Summary of Changes Made
MI	Generate a MAC on an IPB	Can now use an AES keyblock LMK.
PV	Generate a VISA PVV	Can now use an AES keyblock LMK.
R	Load the Diebold Table	The table can now be encrypted using an AES keyblock LMK.
TD	Translate Decimalization Table	Decimalisation tables can now be translated to encryption under an AES keyblock LMK.
V	Verify LMK Store	Now works with AES keyblock LMKs
VC	Verify the Contents of a Smartcard	Can now verify cards holding AES keyblock LMK components.
VT	View LMK Table	Now reports on AES keyblock LMKs

>> Changes to Local HSM Manager menu options to support AES

Managing the HSM

These Local HSM Manager options, used to manage the payShield 9000, have been modified to take account of the AES algorithm.

Menu Option	Summary of Changes Made
File > Login	AES keyblock LMK component cards can now be used to log in.
LMK > Copy LMK Component Card	Can now copy cards holding AES keyblock LMK components.
LMK > Create Authorizing Officer Card	Can now copy cards containing AES keyblock LMK components.
Tools > Health Check and Utilization Statistics > Health Check Data	Tests the AES algorithm
View > HSM Information	Now reports on availability of AES and relevant NIST certifications.

AES Key Management

The following key management HSM Manager options can be used with AES working keys. These commands also support AES keyblock LMKs.

Menu Option	Summary of Changes Made
Keys > Encrypt Key components	Can now encrypt an AES component. TDES components may be, and AES components must be, encrypted under an AES keyblock LMK.
Keys > Export Key	Can now export an AES key. For the key to be exported, TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK. The ZMK used to encrypt the exported keys can be an AES key and/or can be encrypted under an AES keyblock LMK.

Menu Option	Summary of Changes Made
Keys > Form key from components	Can now form an AES key from components. TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK.
Keys > Generate Key	Can now generate an AES key. TDES keys may be, and AES keys must be, encrypted under an AES keyblock LMK. Where the key is to be exported under a ZMK, the ZMK can be an AES key and/or can be protected by an AES keyblock LMK.
Keys > Generate Key components	Can now generate an AES key component. TDES components may be, and AES components must be, encrypted under an AES keyblock LMK.
Keys > Import Key	Can now import an AES key. The ZMK used to encrypt TDES or AES keys to be imported can be an AES key and/or can be encrypted under an AES keyblock LMK. Imported TDES keys may be, and imported AES keys must be, encrypted under an AES keyblock LMK.
Tools > Utilities > Calculate Key Check Value	Can now generate a value for keys, including AES keys, encrypted under an AES keyblock LMK.

AES Keyblock LMK support

The following HSM Manager options are able to accept keys and other data protected by AES keyblock LMKs.

Menu Option	Summary of Changes Made
LMK > Generate LMK	Can now generate AES keyblock LMK components.
LMK > Install Old LMK	Can now work with AES keyblock LMKs. AES Keyblock LMKs cannot be loaded as the old LMK where the new LMK is TDES (either variant or keyblock).
LMK > Uninstall LMK	Can now uninstall AES keyblock LMKs.
LMK > Install LMK	Can now load AES keyblock LMKs.
Tools > Utilities > Calculate CVV	Can now use a CVK encrypted under an AES keyblock LMK.

Menu Option	Summary of Changes Made
Tools > Utilities > Calculate PVV	Can now use an AES keyblock LMK.
Tools > Utilities > Encrypt Decimalisation Table	Decimalisation tables can now be encrypted under an AES keyblock LMK.
Tools > Utilities > Generate MAC on IPB	Can now use an AES keyblock LMK.
Tools > Utilities > Translate Decimalisation Table	Decimalisation tables can now be translated to encryption under an AES keyblock LMK.

>> Appendix A - Abbreviations

AES	Advanced Encryption Standard (algorithm)
BDK	Base Derivation Key (DUKPT)
CAP	Chip Authentication Program
CVK	Card Verification Key
CVV	Card Verification Value
DEA	Data Encryption Algorithm (used for DES)
DES	Data Encryption Standard (algorithm)
DUKPT	Derived Unique Key Per Transaction
HSM	Hardware Security Module
IPB	Issuer Proprietary Bitmap
LMK	Local Master Key
KCV	Key Check Value
PIN	Personal Identification Number
PVV	PIN Verification Value
RSA	Rivest, Shamir, Adelman (algorithm)
TDES	Triple DES (algorithm)
TMK	Terminal Master Key
TR-31	A specification from the ANSI X9 committee for Interoperable Secure Key Exchange Key Blocks for Symmetric Algorithms
ZMK	Zone Master Key

>> Appendix B - Host Command Examples

Below are examples of 2 host commands: for each command, the structure of the command for both TDES Variant LMK and AES Keyblock LMK are given.

For a full description of commands and responses see *[Reference 1](#)*.

Command A0 - Generate a key.

In this example, we want to generate (a) a TDES key to be encrypted under a TDES Variant LMK, and (b) an AES key to be encrypted under an AES Keyblock LMK.

To keep it as simple as possible, this commands requests generation of a key which:

- is a TEK (Terminal Encryption Key)
- is not to be exported (i.e. we are generating the key, but not exporting it under a AMK/TMK at this time).
- is to be encrypted using the default LMK.

Command sent from Host to payShield 9000

Using TDES Variant LMK	
Field	Example or Format
Message Header	1234
Command Code	A0
Mode (e.g. 0 = Generate key)	0
Key Type (e.g. 30B = TEK)	30B
LMK Key Scheme (e.g. U = double-length TDES encrypted using a variant LMK)	U

Using AES Keyblock LMK	
Field	Example or Format
Message Header	1234
Command Code	A0
Mode (e.g. 0 = Generate key)	0
Key Type (Not used - padded with FFF)	FFF
LMK Key Scheme (e.g. S = encrypted using keyblock LMK)	S
Delimiter	#
Key usage (e.g. D0 = Data Encryption Key)	D0
Algorithm (e.g. A3 = 256-bit AES)	A3
Mode of use (e.g. E = may only be used for encryption)	E
Key version number	00
Exportability (e.g. S = controlled by security settings)	S
Number of optional blocks	00

Response sent from payShield 9000 to Host

Using TDES Variant LMK	
Field	Example or Format
Message Header	1234
Response Code	A1
Error Code	00
Key encrypted under the LMK	U+32H
Key Check Value	6H

Using AES Keyblock LMK	
Field	Example or Format
Message Header	1234
Response Code	A1
Error Code	00
Key encrypted under the LMK	S+nA
Key Check Value	6H

Command M0 - Encrypt a block of data.

In this example, we want to encrypt a block of data using (a) a TDES key encrypted under a TDES Variant LMK, and (b) an AES key encrypted under an AES Keyblock LMK.

To keep it as simple as possible, this commands requests encryption of a message which:

- is binary data
- is encrypted in ECB format
- is using an encryption key encrypted under the default LMK.

Command sent from Host to payShield 9000

Using TDES Variant LMK		Using AES Keyblock LMK	
Field	Example or Format	Field	Example or Format
Message Header	1234	Message Header	1234
Command Code	M0	Command Code	M0
Mode (e.g. 00 = ECB)	00	Mode (e.g. 00 = ECB)	00
Input Format Flag (e.g. 0 = Binary)	0	Input Format Flag (e.g. 0 = Binary)	0
Output Format Flag (e.g. 0 = Binary)	0	Output Format Flag (e.g. 0 = Binary)	0
Key Type (e.g. 30B = TEK)	30B	Key Type (Not used - padded with FFF)	FFF
Key, encrypted under the LMK.	U+32H	Key, encrypted under the LMK.	S+nA
Message Length	7H	Message Length	7H
Message to be encrypted	nB	Message to be encrypted	nB

Response sent from payShield 9000 to Host

Using TDES Variant LMK	
Field	Example or Format
Message Header	1234
Response Code	M1
Error Code	00
Message length	<i>4H</i>
Encrypted message	<i>nB</i>

Using AES Keyblock LMK	
Field	Example or Format
Message Header	1234
Response Code	M1
Error Code	00
Message length	<i>4H</i>
Encrypted message	<i>nB</i>

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